

18.1 Master-Slave and Environment Variable Configuration

1. Preparation

- 1) Prepare at least two devices (this lesson uses two MentorPi units as an example).
- 2) Set up the development environment by referring to "**8. Remote Tool Installation and Container Access \ 8.1 VNC Installation and Connection**", and download and install the remote access tool **VNC**.

2. Program Logic

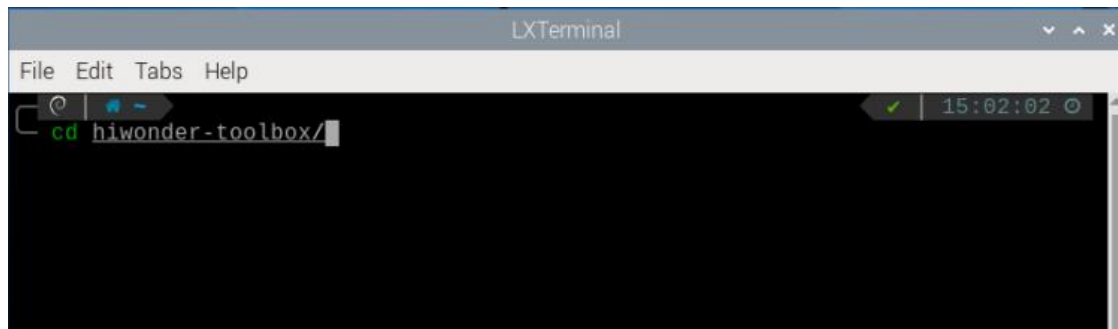
By connecting the master and slave devices to the same network and assigning different namespaces, the broadcast program can send velocity commands to the slave devices, enabling coordinated multi-device control. In this example, a hotspot created by a Raspberry Pi is used for the network connection.

3. Network Configuration

3.1 Configure Host Network

- 1) Power on one MentorPi device and designate it as the master. Once it has booted, remotely access its desktop.
- 2) Open a terminal and navigate to the project directory by entering the following command:

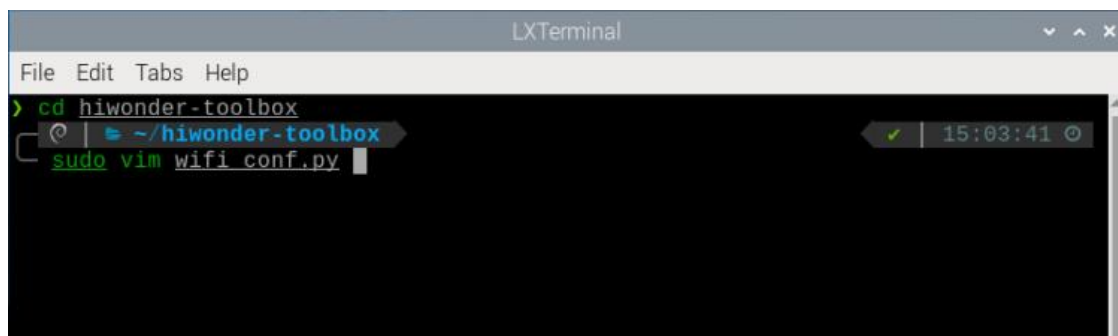
```
cd hiwonder-toolbox
```



```
LXTerminal
File Edit Tabs Help
~ | ~
cd hiwonder-toolbox/
```

3) Open the Wi-Fi configuration file using the Vim editor:

```
sudo vim wifi_conf.py
```



```
LXTerminal
File Edit Tabs Help
> cd hiwonder-toolbox
~ | ~/hiwonder-toolbox
sudo vim wifi_conf.py
```

4) Press the i key to enter edit mode. Locate the two lines of code related to the hotspot password (as shown in the reference image), and update the password as needed.

```
#!/usr/bin/python3
#coding:utf8

WIFI_MODE = 1                                #wifi的工作模式， 1为AP模式， 2为STA
模式
WIFI_AP_SSID = 'HW-123'                      #AP模式下的SSID。字符和数字构成，需要以
HW- 开头，否则app功能无法使用
WIFI_AP_PASSWORD = 'hiwonder'                #AP模式下的WIFI密码，字符和数字构成
#WIFI_STA_SSID = 'abc'                       #STA模式下的SSID
#WIFI_STA_PASSWORD = 'hiwonder'              #STA模式下的WIFI密码

-- INSERT --
```

- 5) After making the changes, press Esc to exit edit mode, then type :wq and press Enter to save and exit the file:

```
:wq
```

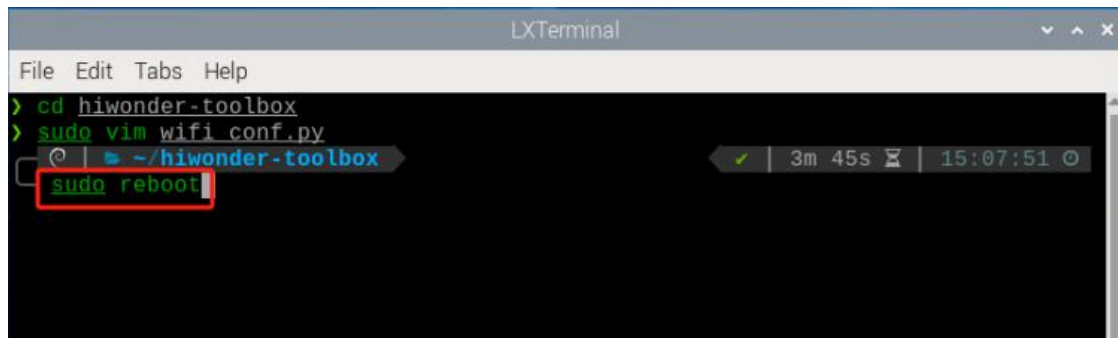
```
#!/usr/bin/python3
#coding:utf8

WIFI_MODE = 1                                #wifi的工作模式， 1为AP模式， 2为STA
模式
WIFI_AP_SSID = 'HW-123'                      #AP模式下的SSID。字符和数字构成，需要以
HW- 开头，否则app功能无法使用
WIFI_AP_PASSWORD = 'hiwonder'                #AP模式下的WIFI密码，字符和数字构成
#WIFI_STA_SSID = 'abc'                       #STA模式下的SSID
#WIFI_STA_PASSWORD = 'hiwonder'              #STA模式下的WIFI密码

:wq
```

- 6) Reboot the master device by running:

```
sudo reboot
```



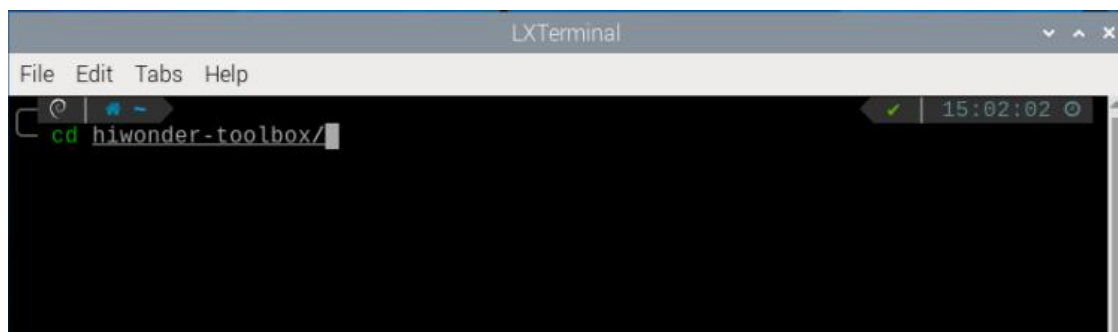
A screenshot of an LXTerminal window. The terminal shows the following commands and output: `cd hiwonder-toolbox`, `sudo vim wifi_conf.py`, and `sudo reboot`. The `sudo reboot` command is highlighted with a red rectangle. The terminal also shows a prompt `~ /hiwonder-toolbox` and a status bar with a green checkmark, '3m 45s', and the time '15:07:51'.

- 7) Once the device has restarted, connect to the master's Wi-Fi using the updated password. The default password is "hiwonder" unless you specified a different one during configuration.

3.2 Configure Slave Network

- 1) Power on the slave device and remotely access its desktop environment.
- 2) Open a terminal and navigate to the project directory with the following command:

```
cd hiwonder-toolbox
```



A screenshot of an LXTerminal window. The terminal shows the command `cd hiwonder-toolbox/` being entered. The terminal also shows a prompt `~` and a status bar with a green checkmark and the time '15:02:02'.

- 3) Use the Vim editor to open the Wi-Fi configuration file:

```
sudo vim wifi_conf.py
```

```

LXTerminal
File Edit Tabs Help
> cd hiwonder-toolbox
~ | ~/hiwonder-toolbox
sudo vim wifi_conf.py
15:03:41

```

- 4) Press **i** to enter **edit mode**. Locate the three lines of code shown in the reference image and modify the Wi-Fi name and password as indicated.

Located the indicated codes.

```

#!/usr/bin/python3
#coding:utf8

WIFI_MODE = 2 #wifi的工作模式， 1为AP模式， 2为STA模式
#WIFI_AP_SSID = 'HW-Robot' #AP模式下的SSID。字符和数字构成，需要以 HW- 开头，否则app功能无法使用
#WIFI_AP_PASSWORD = 'hiwonder' #AP模式下的WIFI密码，字符和数字构成
WIFI_STA_SSID = 'HW-123' #STA模式下的SSID
WIFI_STA_PASSWORD = 'hiwonder' #STA模式下的WIFI密码

-- INSERT --
7,24 All

```

Set the **network mode** to “**2**” to enable **LAN mode**.

Use “**HW-123**” as the Wi-Fi name and “**hiwonder**” as the password—these should match the hotspot settings configured on the master device in **Section 3.1**.

- 5) Once editing is complete, press **Esc** to exit edit mode, then type **:wq** and press **Enter** to save and close the file:

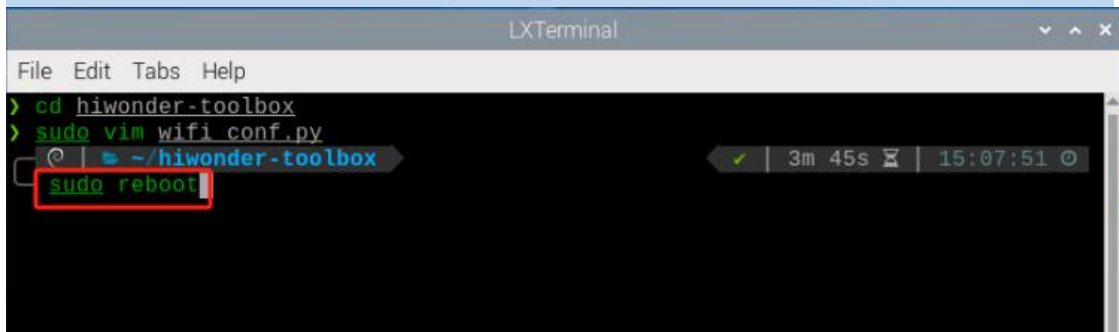
```
:wq
```

```
#!/usr/bin/python3
#coding:utf8

WIFI_MODE = 2                                #wifi的工作模式， 1为AP模式， 2为STA
模式                                         #AP模式下的SSID。字符和数字构成，需要
#WIFI_AP_SSID = 'HW-Robot'                  #AP模式下的WIFI密码，字符和数字构成
以 HW- 开头，否则app功能无法使用          #STA模式下的SSID
#WIFI_AP_PASSWORD = 'hiwonder'              #STA模式下的WIFI密码
WIFI_STA_SSID = 'HW-123'
WIFI_STA_PASSWORD = 'hiwonder'
```

6) Reboot the device by entering the following command (this step is required):

sudo reboot



```
LXTerminal
File Edit Tabs Help
> cd hiwonder-toolbox
> sudo vim wifi_conf.py
~/.hiwonder-toolbox
sudo reboot
```

Note:

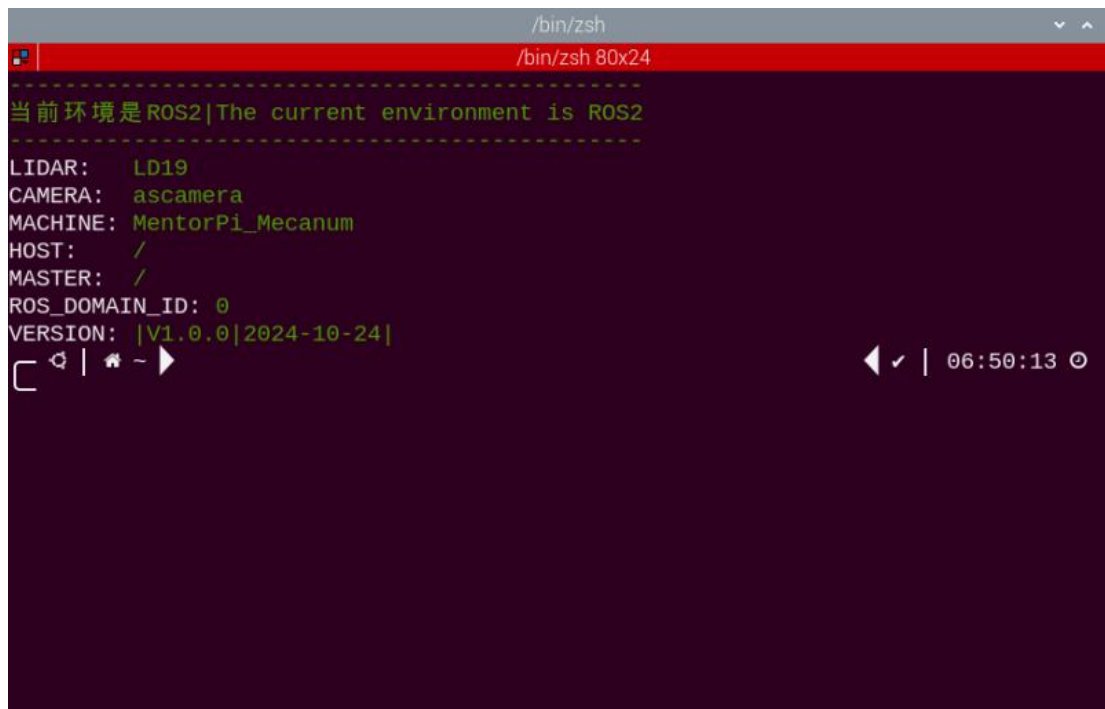
- ① When MentorPi starts in **LAN mode**, it will first attempt to automatically connect to the predefined network. During this process, **LED2 on the Raspberry Pi expansion board will flash rapidly**, indicating that the device is actively searching for the network.
- ② If the target network cannot be found after **three consecutive search attempts**, MentorPi will automatically switch to **Direct Connection mode** (indicated by **LED2 flashing slowly**).

③ Please note that when MentorPi switches back to Direct Connection mode due to a failed LAN connection, the settings in the `wifi_conf.py` file **will not be updated automatically**—they will still reflect the LAN configuration. Therefore, unless the file is manually modified, MentorPi will continue to attempt LAN mode first on the next startup.

4. Configure Environment Variable

4.1 Configuring Environment Variables on the Master Device

- 1) Select one MentorPi vehicle as the master and power it on. Use VNC to remotely connect to the Raspberry Pi desktop.
- 2) Click  in the top-left corner of the desktop to open the Terminator terminal.
- 3) Ensure the master device is connected to the USB game controller receiver.



```
/bin/zsh
/bin/zsh 80x24
-----
当前环境是ROS2|The current environment is ROS2
-----
LIDAR:   LD19
CAMERA:  ascamera
MACHINE: MentorPi_Mecanum
HOST:    /
MASTER:  /
ROS_DOMAIN_ID: 0
VERSION: |V1.0.0|2024-10-24|
[ 06:50:13 ]
```

- 4) Enter the command to disable automatic startup of the gameplay and

press Enter:

`~/stop_ros.sh`

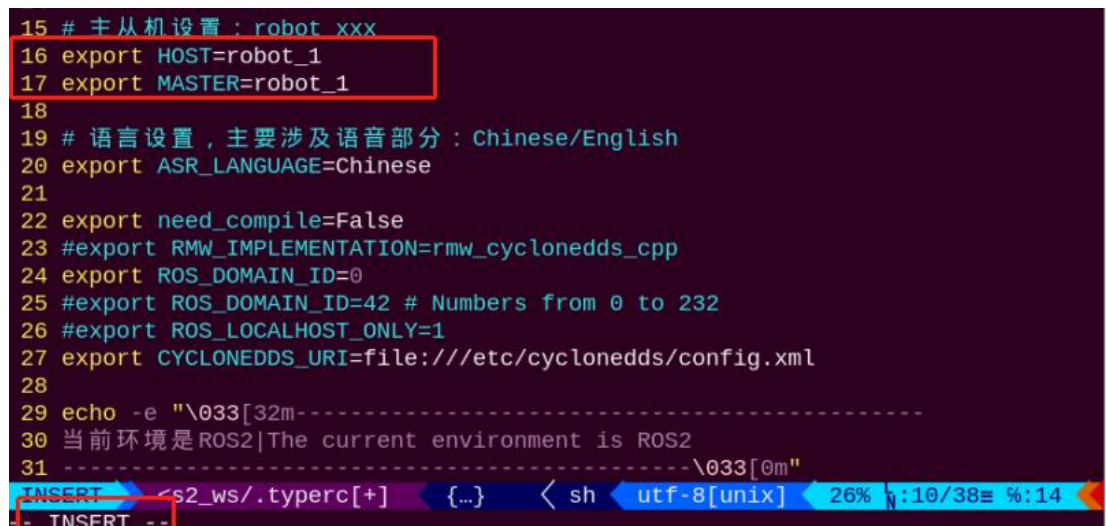
```
-----
当前环境是ROS2|The current environment is ROS2
-----
LIDAR: LD19
CAMERA: ascamera
MACHINE: MentorPi_Mecanum
HOST: /
MASTER: /
ROS_DOMAIN_ID: 0
VERSION: |V1.0.0|2024-10-24|
> ~/stop_ros.sh
zsh: killed ~/stop_ros.sh
```

- 5) Open the .typerc file to configure environment variables for the master device by entering:

`vim /home/ubuntu/ros2_ws/.typerc`



- 6) Press the i key to enter edit mode. Enable the master-slave settings by setting the master to robot_1/.



```
15 # 主从机设置 : robot xxx
16 export HOST=robot_1
17 export MASTER=robot_1
18
19 # 语言设置 , 主要涉及语音部分 : Chinese/English
20 export ASR_LANGUAGE=Chinese
21
22 export need_compile=False
23 #export RMW_IMPLEMENTATION=rmw_cyclonedds_cpp
24 export ROS_DOMAIN_ID=0
25 #export ROS_DOMAIN_ID=42 # Numbers from 0 to 232
26 #export ROS_LOCALHOST_ONLY=1
27 export CYCLONEDDS_URI=file:///etc/cyclonedds/config.xml
28
29 echo -e "\033[32m-----
30 当前环境是ROS2|The current environment is ROS2
31 -----\033[0m"
INSERT ~/ros2_ws/.typerc[+] {...} < sh utf-8[unix] 26% 10/38 %:14
- INSERT -
```

- 7) After making the changes, press Esc, then type :wq and press Enter to save and exit.


```

13 export MACHINE_TYPE=MentorPi_Mecanum
14
15 # 主从机设置 : robot_xxx
16 export HOST=robot_1
17 export MASTER=robot_1
18
19 # 语言设置 , 主要涉及语音部分 : Chinese/English
20 export ASR_LANGUAGE=Chinese
21
22 export need_compile=False
23 #export RMW_IMPLEMENTATION=rmw_cyclonedds_cpp
24 export ROS_DOMAIN_ID=0
25 #export ROS_DOMAIN_ID=42 # Numbers from 0 to 232
26 #export ROS_LOCALHOST_ONLY=1
COMMAND -z_ws/.typerc[+] {...} ON < sh utf-8[unix] 44% N:17/38≡ %:21
:wq

```

8) Finally, enter the following command to apply the master configuration:

source ~/.zshrc

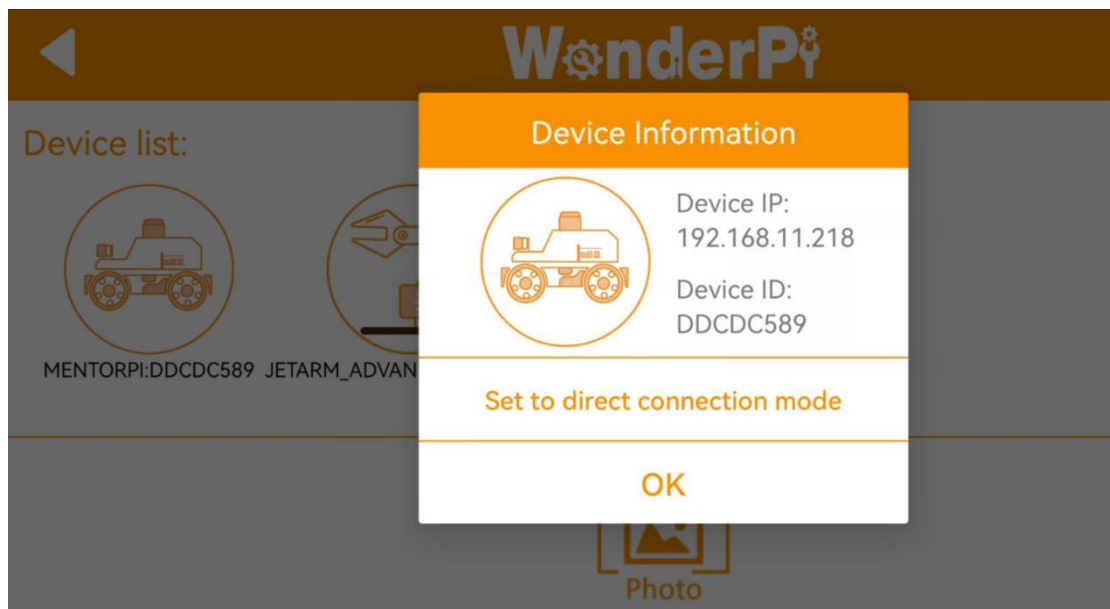
```

> source ~/.zshrc
-----
当前环境是ROS2|The current environment is ROS2
-----
LIDAR: LD19
CAMERA: ascamera
MACHINE: MentorPi_Mecanum
HOST: robot_1
MASTER: robot_1
ROS_DOMAIN_ID: 0
VERSION: |V1.0.0|2024-10-24|

```

4.2 Configuring Environment Variables on the Slave Device

- 1) Connect your mobile app to the Wi-Fi network generated by the master device. Perform a search, and you will see two car icons displayed. Long-press the icon representing the slave device to reveal its IP address. Use this IP to follow the instructions in “8. Remote Tool Installation and Container Access \ 8.1 VNC Installation and Connection”.



- 2) After accessing the device, follow the steps in “**4.1 Configuring Environment Variables on the Master Device**” First, open the .typerc file and set the slave device’s environment variable **HOST** to robot_2. Then save and exit the file. Finally, run the command `source ~/.zshrc` to refresh the environment variables.

```
6 # 雷达类型 : A1/A2/C1/G4/S2L/LD14P/MS200
7 export LIDAR_TYPE=LD19
8
9 # 深度摄像头类型 : AstraPro/AstraProPlus/None
10 export DEPTH_CAMERA_TYPE=ascamera
11
12 # 机器类型 : JetAcker
13 export MACHINE_TYPE=MentorPi_Mecanum
14
15 # 主从机设置 : robot_xxx
16 export HOST=robot_2
17 export MASTER=robot_1
18
19 # 语言设置, 主要涉及语音部分 : Chinese/English
20 export ASR_LANGUAGE=Chinese
21
22 export need_compile=False
23 #export RMW_IMPLEMENTATION=rmw_cyclonedds_cpp
24 export ROS_DOMAIN_ID=0
25 #export ROS_DOMAIN_ID=42 # Numbers from 0 to 232
```

```
> source ~/.zshrc
-----
当前环境是ROS2|The current environment is ROS2
-----
LIDAR:    LD19
CAMERA:   ascamera
MACHINE:  MentorPi_Mecanum
HOST:     robot_2
MASTER:   robot_1
ROS_DOMAIN_ID: 0
VERSION:  |V1.0.0|2024-10-24|
```